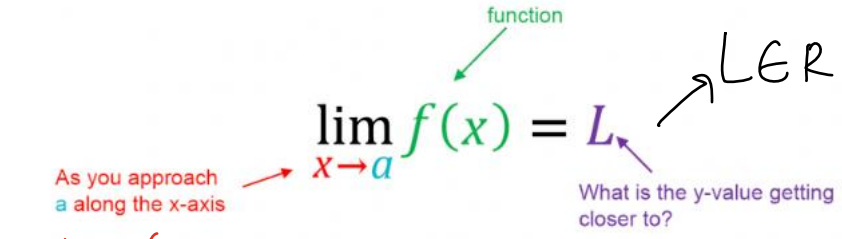
### 2.1.1 Definition of limits




$L \in \mathbb{R}$

$x \to a$ ✓

$x = a$ ✗

(1) The right-hand limit of function $f(x)$ as x approaches a is written as $\lim_{x \to a^+} f(x)$.

$x \to a^+$

(2) The left-hand limit of function $f(x)$ as x approaches a is written as $\lim_{x \to a^-} f(x)$.

$x \to a^-$

Both are known as the one-sided limits.

- right limit
- left limit

(3) The two sided limit $\lim_{x \to a} f(x)$ ✓ exist if and only $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = L$ where $L \in \Re$.

The two-sided limit exist if the one-sided limits are the same.

Techniques:

① Substitute

② If $\frac{0}{0}$ or $\frac{\infty}{\infty}$:

(a) factorize

③ If $x \to -\infty$ or $x \to +\infty$, take the highest term (of f(x)).